

COLLEGE BIOMETRIC ATTENDANCE SYSTEM

Technical Summary: Reverse Engineering, Custom Development & Modern UI Architecture

Target Terminal: eSSL / ZKTeco Standalone Reader	Terminal IP: 192.168.1.201 : 4370
Comm Key / Password: 0 (Standard Authenticated)	Replaced Application: eTimeTrackLite (Windows)
Status: 100% License-Free • Fully Functional	Date: October 05, 2026

1. Executive Summary

This project successfully reversed the proprietary Windows **eTimeTrackLite** attendance software and engineered a modern, license-free, in-house replacement designed specifically for our college. The newly developed system communicates directly with the biometric reader on the college LAN (192.168.1.201:4370), stores attendance data in an offline SQLite database, calculates daily staff working hours, and exports publication-ready PDF and CSV reports. The interface has been fully upgraded to a 2026 minimalist, glassy white enterprise dashboard featuring sidebar navigation and a live connection health indicator.

2. Reverse Engineering Technical Audit

Static and architectural analysis of the original Windows application package revealed the exact protocol, runtime dependencies, and communication mechanisms used by eTimeTrackLite:

Category	Technical Discovery & Evidence
Installer Package	Embedded MSCF cabinet inside eTimeTrackLite.msi at offset 8,302,080. Extracted 203 original application assemblies and assets.
Binary Framework	VB.NET targeting Microsoft .NET Framework 4.0 (x86) . Compiled with Visual Studio 2010. Native WinForms frontend.
Biometric SDK	ZKTeco Standalone SDK (zkemkeeper.dll v6.3.1.55) . Backed by C/C++ native drivers tcpcomm.dll and pltcpcomm.dll.
Transport Protocol	Direct TCP stream on hardcoded port 4370 . Framed by ZKTeco magic bytes 0x50 0x50 0x82 0x7D followed by command headers.
Authentication	CommKey = 0. Handshake executes CMD_CONNECT (1000) followed by CMD_AUTH (1102) key hash verification.
Log Retrieval	Read-only execution of CMD_ATTLOG_RRQ (13). Reads all punch buffers without resetting the device memory.

3. In-House Software Architecture

To replace eTimeTrackLite cleanly without licensing fees or Windows lock-in, we developed a modular Python application residing in college_attendance_app/:

Module	Role & Core Implementation
app.py	Full GUI client. Features a modern left sidebar, top app bar with live status pill, dynamic KPI cards, and centered modal dialogs for Staff CRUD.
device_driver.py	Dual-layer hardware communication engine. Uses pyzk for high-level operations with pure Python raw socket fallback (TCP 4370).
database.py	Embedded SQLite manager (college_attendance.db). Manages tables for staff, raw immutable punch logs, processed daily attendance, and hardware settings.
attendance_engine.py	Automated attendance rules engine. Calculates First IN, Last OUT, total work duration, and statuses: <i>Present, Half Day, Short Leave, Missed Out, Absent</i> .
pdf_generator.py	Pure ReportLab PDF generator. Produces multi-page landscape reports with college header, KPI scorecard, attendance grid, and signature approval blocks.
run_app.py	Intelligent root launcher. Automatically detects if the active Python lacks Tkinter and re-routes to a compatible Python runtime seamlessly.

4. Custom College Features & Refinements

A. Full 12 College Departments Integration

The system has been configured with the exact departmental structure of our college across all search bars, dropdown filters, database tables, and PDF reports:

1. AI & DS • 2. CSE • 3. IT • 4. ECE • 5. EEE • 6. MECH • 7. Physics • 8. Chemistry • 9. English • 10. Administration • 11. Library • 12. Maintenance

B. Hardware & Access Control Explained

- **Device Administrator Role (privilege = 14):** Locks the physical keypad on the biometric reader. When enabled, unauthorized persons cannot press the M/OK button to access device menus or alter network configurations. Only verified Admins can unlock the machine.
- **Direct EEPROM Sync (Push to Terminal):** Immediately transfers new staff registrations to the biometric terminal's onboard memory chip. When the staff member scans their fingerprint, the machine displays their name (e.g. 'Welcome, Dr. Rajesh').
- **Communication Key (CommKey = 0):** The hardware security password preventing unauthorized network machines from accessing biometric records. Set to manufacturer default 0, matching our college hardware configuration.

C. Modern Glassy White UI & Live Connectivity Pill

- **Zero Gradient, Minimalist White Theme:** Crisp white surfaces (#FFFFFF), slate canvas (#F8FAFC), and 1px subtle divider lines (#E2E8F0). No beveled 1990s borders.
- **Automated Green / Red Connection Heartbeat:** On launch, the top app bar asynchronously probes the device. Turns **GREEN** (■ Online: 192.168.1.201:4370) when on the college LAN, and **RED** (■ Offline: 192.168.1.201:4370) when working from home. Re-checks automatically every 30 seconds.
- **Soft Pastel Status Badges:** Attendance grid tags rows with color-coded badges: Green for Present, Amber for Half Day, Blue for Missed Out, and Red for Absent.

5. Comparison: eTimeTrackLite vs. In-House System

Feature / Capability	eTimeTrackLite (Legacy)	In-House System (Our Solution)
License & Cost	Paid / Dongle License required	100% Free & Open Source
Platform Support	Windows only (x86 .NET 4.0)	Cross-Platform (macOS & Windows)
Database	MS Access / SQL Server	Embedded SQLite (Zero setup)
UI Design	Legacy 2005 Windows Forms	2026 Glassy White Dashboard
Live Hardware Status	Manual sync click only	Auto 30s Green/Red Heartbeat
Report Output	Proprietary Crystal Reports	Clean PDF (Printable) & CSV

6. Deployment & Standalone Compilation

The application is completely self-contained. To run or deploy:

- **Running on Mac or Linux:**

`python3 run_app.py` (Automatically handles Tkinter environment detection).

- **Compiling to a Standalone Windows .exe:**

On any Windows machine with Python installed, run:

`pip install pyzk reportlab openpyxl pyinstaller`

`pyinstaller --noconsole --onefile --name="CollegeAttendanceSystem" run_app.py`

The standalone single-click executable will be created in the `dist/` folder.

Prepared By: Antigravity Systems
Agent

Verified For: College Administration

Project Status: Production Ready